

Engineering conclusion

Yes—adding a central relief between two guide lands is the better design for hot-runner maintenance, provided that the relieved area is protected from PET ingress.

For the original full-length close-fit design, any contact caused by thermal expansion, contamination, burrs, distortion or PET deposits can occur over the complete engagement length:

$$F_{\text{pull, full}} = \mu p \pi D L_T$$

With the relieved design shown, only the front and rear guide lands are potential contact areas:

$$F_{\text{pull, relieved}} = \mu p \pi D (L_1 + L_2)$$

Therefore, the approximate ratio of frictional pull forces is:

$$\boxed{\frac{F_{\text{relieved}}}{F_{\text{full}}} \approx \frac{L_1 + L_2}{L_T}}$$

and the theoretical reduction is:

$$\boxed{\text{Force reduction} = \left[1 - \frac{L_1 + L_2}{L_T} \right] \times 100\%}$$

For example, if the original engagement length is 60 mm and the two guide lands total 20 mm:

$$\text{Reduction} = \left(1 - \frac{20}{60} \right) \times 100 \approx 67\%$$

This means the friction-related extraction force could be approximately one-third of the full-contact design, assuming the same contact pressure and friction coefficient.

Fresh hot runner

Because the insert is already **10 µm smaller than the hole diametrically**, the ideal theoretical fit force is nearly zero when:

- Both components are at approximately the same temperature.
- The bore and insert are clean.
- There is no distortion or misalignment.
- No PET or other contamination is present.

In the fresh condition, the relief may therefore not produce a large measurable reduction in ideal insertion force. Its main advantage is that it makes the design more tolerant of actual manufacturing and maintenance conditions.

Only the two lands need to be accurately ground, inspected and cleaned. Minor deposits, thermal distortion or surface damage in the middle region will not directly contact the bore.

Used hot runner

The maintenance advantage becomes more important after service because the relief:

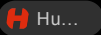
- Reduces the length over which PET residue can create friction.
- Reduces steel-to-steel rubbing and scoring.
- Reduces the possibility of galling during extraction.
- Limits polishing and dimensional restoration to two short guide lands.
- Provides better tolerance against small bore distortions.
- Makes it easier to break the insert free during maintenance.

The two separated lands can still provide good radial and angular guidance, provided they are coaxial and have sufficient land length.

Critical PET condition

The conclusion is valid only when PET cannot enter and solidify inside the relieved annular area.

If molten PET passes the first guide land and fills the relief, it may form a solid annular ring after cooling. That ring can act like a mechanical lock, potentially making the relieved insert more difficult to remove than a straight insert.

OEM hot-runner maintenance guidance similarly emphasizes clean and burr-free fitting diameters, checking for interference instead of forcing components, relieving residual pressure, and minimizing molten-plastic entry into temporary internal voids. 

Therefore, the design should include:

1. A sufficient sealing or guiding land between the melt-pressure region and the relief.
2. No direct connection between the relief and the PET flow channel.
3. A relief depth substantially greater than the 10 μm fit clearance, so normal deposits or thermal movement cannot bridge the relief.
4. Smooth transitions or radii at the land-to-relief shoulders.
5. Lead-in chamfers at both guiding lands.
6. An extraction thread, puller groove or jacking arrangement for controlled removal.
7. A defined maintenance procedure for depressurization and hot-versus-cold removal.

Report-ready conclusion

The insert with a central relieved section and two separated guiding lands is preferred over a full-length close-fitting insert for hot-runner maintenance. The relief reduces the effective contact length and consequently reduces the potential insertion and extraction friction when thermal expansion, contamination, surface deposits or minor misalignment create radial contact. It also limits precision fitting, inspection, cleaning and polishing to the two guide lands, thereby reducing maintenance effort and the risk of scoring or seizure. The benefit is conditional upon preventing molten PET from entering the relieved area, because solidified PET inside the relief could form an annular mechanical lock. Subject to proper melt sealing, adequate relief depth, coaxial guide lands and a suitable extraction feature, the relieved insert is the more maintainable design.

A more concise drawing-note version is:

Central relief provided to reduce effective bearing/contact length, minimize extraction force and facilitate hot-runner maintenance. Relief shall remain isolated from the PET melt path to prevent polymer locking.